NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date: 05.01.2025
Time: 3HRS SPCL_GTM-01 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
C	C	A	B	C	B	D	C	A	D
11	12	13	14	15	16	17	18	19	20
B	D	A	A	B	A	C	A	C	C
21	22	23	24	25	26	27	28	29	30
3	6	49	7	2					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	C	D	A	A	A	C	C	D
36	37	38	39	40	41	42	43	44	45
B	C	D	C	A	D	C	C	D	C
46	47	48	49	50					
84	1	3	4	960					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	B	C	D	B	A	B	C	C	B
61	62	63	64	65	66	67	68	69	70
A	B	C	D	A	B	D	D	A	A
71	72	73	74	75					
4	15	12	10	8					

APEX JEE-MAIN 2025

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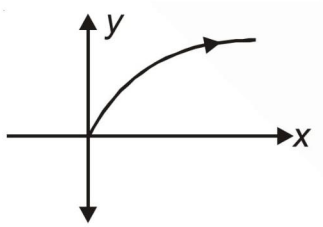
HINTS&SOL
PHYSICS

1. Ray becomes parallel to base for minimum deviation $\frac{\sin\theta}{\sin r}=\sqrt{2}\Rightarrow r=30$

   Hence $\theta=45^0$

2. The tension in the string is responsible to give centripetal acceleration to the center of mass of rod. Thus,

   $$T=M\omega^2\left(\frac{3L}{2}\right)$$

3. 

   $x=\frac{t^2}{2}\quad y^2=2\times\frac{t^2}{2}$

   $V_x=t\quad y=t$

   $a_x=1\quad a_y=0$

4. $PV=nRT=nR\times\frac{\sqrt{\alpha}}{\sqrt{P}}$

   $\Rightarrow P^{3/2}V=C\Rightarrow\frac{3}{2}VP^{1/2}\frac{dP}{dV}+P^{3/2}=0$

   $\Rightarrow\frac{3}{2}V\frac{dP}{dV}=-P\Rightarrow -V\frac{dP}{dV}=\frac{2}{3}P$

5. Threshold wavelength $\left(\lambda_0\right)=\frac{c}{f_0}$

   Since, $f_k<f_{Be}$

   $\Rightarrow\lambda_k>\lambda_{Be}$

6. The emf is given as:

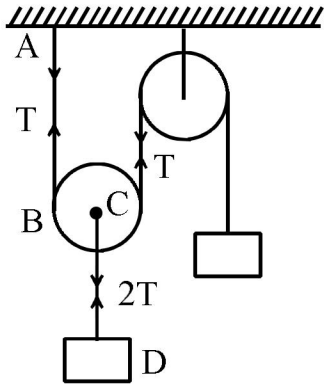
   $$\varepsilon=\left(\vec{B}\times\vec{v}\right).\left(\vec{l}\right)$$

   $$\varepsilon=\left[\left(3\hat{i}+4\hat{j}+5\hat{k}\right)\times\left(\hat{i}\right)\right].\left(5\hat{j}\right)$$

   $$\varepsilon=25\text{V}$$

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7. $V_1 = \sqrt{\frac{T}{\mu}}$

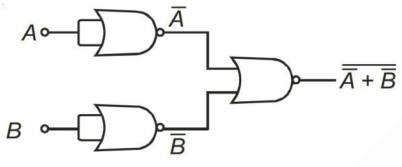


$V_2 = \sqrt{\frac{2\%}{\mu}}, \frac{V_1}{V_2} = \frac{1}{\sqrt{2}}$

8. $\gamma = \frac{5}{3}$

$\therefore \quad x = \frac{5}{3} - 1 = \frac{2}{3}$

9.



From De-Morgan's law

$\overline{\bar{A}+\bar{B}} = \bar{\bar{A}}.\bar{\bar{B}} = A.B$

10. Conceptual

11. Given $E_3 - E_2 = \frac{hc}{\lambda_1}$

$E_2 - E_1 = \frac{hc}{\lambda_2}$

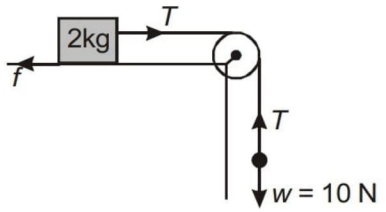
$E_3 - E_1 = \frac{hc}{\lambda_3}$

$\Rightarrow \frac{hc}{\lambda_3} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_1} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

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12. $V=-\frac{Gm}{1}-\frac{Gm}{2}-\frac{Gm}{3}-\ldots=-Gm\left(1+\frac{1}{2}+\frac{1}{4}+\ldots\right)=-2Gm$

13.



System will not moves as $f_{\max}=0.6\times2\times10>1\times10(weight)$

$\therefore T=w=10N$

14. $\frac{\Delta P_1}{\Delta P_2}=\frac{0.01}{0.02}=\frac{1}{2}$

or $\frac{4T/r_1}{4T/r_2}=\frac{1}{2}$ (or) $\frac{r_1}{r_2}=2$

$\therefore \frac{V_1}{V_2}=\left(\frac{r_1}{r_2}\right)^3=8$

15. One resistance of 2Ω is short circuited

16. The system can be considered as group of two dipoles, with dipole moments

$\overrightarrow{p_1}=-Qx\hat{i}$ and $\overrightarrow{p_2}=-Qx\hat{j}$

Hence, $\vec{p}=-Qx\left(\hat{i}+\hat{j}\right)$

17. $\frac{CV^2}{\varepsilon_0E^2}=\frac{energy}{energy/volume}=volume$

$\frac{mass}{\text{density}}=volume$

18. The magnetic field will be maximum at the surface of the cylindrical wire

$$B=\begin{cases}\frac{\mu_0Ir}{2\pi R^2}, 0\le r\le R\\ \frac{\mu_0I}{2\pi r},\ r\ge R\end{cases}$$

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$$B_{\max imum} = \frac{\mu_0 I}{2\pi R^2} \times R = \frac{\mu_0 I}{2\pi R}$$

19. $h = u\sin\theta t - \frac{1}{2}gt^2 \Rightarrow t_1 t_2 = \frac{2h}{g}$

20. Using momentum conservation along horizontal direction:

$$2MV_1\cos 30° + MV_2\cos 45° = M\left(10\text{m/s}\right)\cos 30° + 2M\left(5\text{m/s}\right)\cos 45°$$

$$\Rightarrow \quad \sqrt{3}v_1 + \frac{v_2}{\sqrt{2}} = 5\sqrt{3} + 5\sqrt{2}$$

Using momentum conservation along vertical direction:

$$2MV_1\sin 30° - MV_2\sin 45° = -M\left(10\text{m/s}\right)\sin 30° + 2M\left(5\text{m/s}\right)\sin 45°$$

$$\Rightarrow \quad v_1 - \frac{v_2}{\sqrt{2}} = -5 + 5\sqrt{2}$$

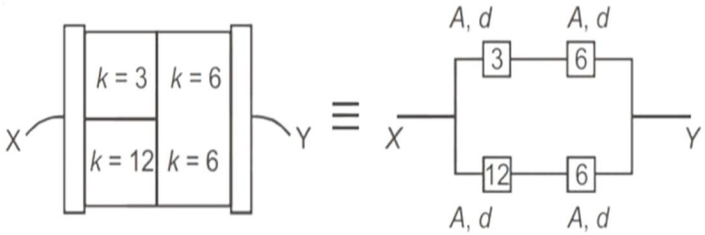
Solving, we get:

$$v_1 = \frac{5\left(\sqrt{3}-1\right)+10\sqrt{2}}{\sqrt{3}+1}$$

$$\Rightarrow \quad v_1 = \left(\frac{5\left(0.732\right)+14.14}{2}\right)\left(0.732\right)\text{m/s} = 6.515\text{m/s}$$

21. Specify gravity or relative density $= \frac{\text{weight in air}}{\text{change in weight in water}} = \frac{60}{20} = 3$

22.



Equivalent capacitance of the circuit $\frac{k\varepsilon_0\left(2A\right)}{\left(2d\right)} = \frac{3\times 6}{3+6}.\frac{\varepsilon_0 A}{d} + \frac{6\times 12}{6+12}.\frac{\varepsilon_0 A}{d}$

$\Rightarrow k = 2 + 4 = 6$

23. $\frac{I_{\max}}{I_{\max}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \left(\frac{\sqrt{16I}+\sqrt{9I}^2}{\sqrt{16I}-\sqrt{9I}}\right) = 49$

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24. $mgh = \frac{1}{2} \times \left(\frac{7}{5} mr^2\right) \times \left(\frac{v_0^2}{r^2}\right) \quad \Rightarrow h = \frac{7}{10}\frac{v_0^2}{g}$

25. $\phi = MI = \frac{\mu_0 I}{2R} l^2 \quad \Rightarrow M = \frac{\mu_0 l^2}{2R}$

CHEMISTRY

26. Conceptual

27. Time period $\propto \frac{n^3}{z^3}$

$$\frac{T_1}{T_2} = \frac{n_1^3 \times z_2^3}{n_2^3 \times z_1^3}; z_1 = z_2$$

Hence, $\frac{T_1}{T_2} = \frac{1^3}{3^3} = \frac{1}{27}$

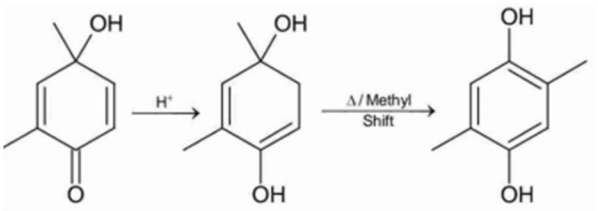
28. From Keq value; % of β-D-glucose is 64.3

$$\underset{1-x}{\alpha - D} \rightleftharpoons \underset{x}{\beta - D}$$

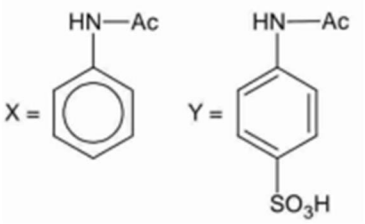
$$K_C = \frac{x}{1-x} = 1.8 \Rightarrow x = 0.643$$

$$112 \times (1 - 0.643) + 0.643 \times \phi = 59^0$$

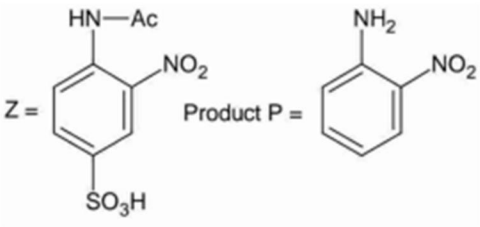
$$\phi = specific\, rotation\, of\ \beta - D\ = +29.6^0$$

29. 

30. All others are used for this purpose and they form cyclic acetal or cyclic thioacetals at  group.

31. 

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32. $[Ni(CN)_4]^{2-} \Leftarrow dsp^2$, square planer and diamagnetic
$[Cu(NH_3)_4]^{2+} \Leftarrow dsp^2$, square planer and paramagnetic
$[Ni(CO)_4] \Leftarrow$ is sp^3, tetrahedral but diamagnetic

33. C)

$$O=\underset{\underset{O}{\|}}{Br}-\underset{\underset{O}{\|}}{Br}-\underset{\underset{O}{\|}}{Br}=0$$

34. For a diatomic gas like N_2

$$\frac{\Delta H}{\Delta U}=\frac{C_P}{C_V}=\gamma=\frac{5}{3}$$

35. In reaction D, no change in oxidation state of any element
36. Only B contains chiral centre
37. This is a 3^0 and resonance stabilized carbocation.
38. Cumulated diene is least stable, evolve maximum heat in the hydrogenation reaction.
39. Vapour pressure depends on temperature not on size of container.
40. $3A + B \rightarrow 2C$

$$r=\frac{1}{3}\frac{d[A]}{dt}=-\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}=K'[A][B]$$

$$-\frac{d[A]}{dt}=3K'[A][B]=K[A][B]$$

i.e., K = 3K'

So that $\frac{d[C]}{dt}=2K'[A][B]=\frac{2}{3}K[A][B]$

41. $K_P = P_{CO_2}$ which will depends temperature of the system only
42. Water will be neutral as $[H^+] = OH^-]$

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43. $E_{cell} = E^0_{cell} - \frac{0.06}{n}\log\frac{\left[Cr^{3+}\right]^2}{\left[Cr_2O_7^{2-}\right]\left[H^+\right]^8}$

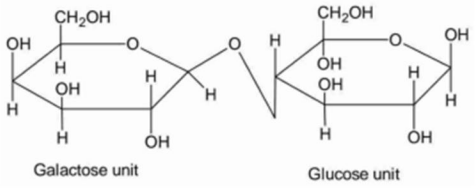
$1.05V = 1.33 - \frac{0.06}{6}\log\frac{(0.1)^2}{0.01\times\left[H^+\right]8}$

$0.01\log\frac{1}{\left[H^+\right]^{14}} = 0.28$

$-0.14\ \log\left[H^+\right]^8 = 0.28$

$-\log\left[H^+\right] = \frac{0.28}{0.14} = 2$

44.



45. Conceptual

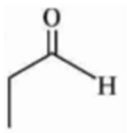
46. $3\times moles\ of\ C_2H_5OH = 3\times mole\ of\ C_2H_4O_2$

moles of $C_2H_5OH = \frac{5}{46}(Theoretical)$

moles of $C_2H_4O_2$ (Experimental) $= \frac{5.5}{60}$

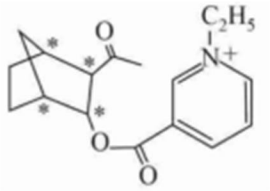
% yield $= \frac{5.5}{60\times5}\times46\times100 = 84\%$

47.

Only 

48. ClF_3, NH_3 and H_2O are polar

49.



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50.

$$\text{Moles of ethanol} = \frac{460}{46} = 10$$

$$\text{Moles of methanol} = \frac{x}{32}$$

$$\text{Moles fraction of ethanol} = \frac{10}{10+\frac{x}{32}} = \frac{320}{320+x}$$

$$\text{Vapour pressure of solution} = P_{\text{ethanol}} + P_{\text{methanol}}$$

$$P_T = P_A^o x_A + P_B^o x_B$$

$$72 = \left(\frac{320}{(320+x)} \times 48\right) + \left(\frac{x}{(320+x)} \times 80\right)$$

$$72 = \frac{(320\times 48)+80x}{(320+x)}$$

$$x = 960$$

MATHEMATICS

51. Distance b/w two lines = 8

Radius = 4

Le equation of one of diameter be $3x+4y+t=0 \rightarrow (1)$

$2x+y=0 \rightarrow (2)$

Solve $(1)\&(2) \Rightarrow (x, y) = \left(\frac{t}{5}, -\frac{2t}{5}\right)$

R = distance $\left(\frac{t}{5}, -\frac{2t}{5}\right)$ from $4x-3y+10=0$

$$4 = \left|\frac{\frac{4t}{5}+\frac{6t}{5}+10}{5}\right| \Rightarrow t = 5, -15$$

Case – i) $t=5$, centre $=(1,-2)$

$(x-1)^2+(y+2)^2 = 4^2 \Rightarrow X-\text{intercept} = 2\sqrt{g^2-c} = 2\sqrt{12}$

Square of it $= 4\times 12 = 48$

52. $V = npq; n=10, p=\frac{3}{5}, q=\frac{2}{5}$

53. $f(c)<0; f(d)>0$

54. $ABCD$ is a parallelogram $\Rightarrow D = A+C-B$

$\Rightarrow \alpha = 3; \beta = 21$

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$\alpha x+\beta y=k$ is passing through $(8,11)\Rightarrow k=255$

55. $\sin x-2<0\forall x\in R\ \&\ x^2+x+2>0\forall x\in R$

$$\Rightarrow\frac{\left(e^x-1\right)(2x-3)}{x(x+1)}\geq 0$$

Case – i) $e^x-1>0\Rightarrow x>0\Rightarrow x=\left[\frac{3}{2},\alpha\right)\rightarrow(1)$

Case – ii) $e^x-1<0\Rightarrow x<0\Rightarrow\frac{2x-3}{x(x+1)}\leq 0$

$x<0\Rightarrow x\in(-\alpha,-1)\rightarrow(2)$

From $(1)\&(2)\ x\in(-\infty,-1)\cup\left[\frac{3}{2},\infty\right)$

56. $f(x)=-1+|x-2|,0\leq x\leq 4$

$$f(x)=\begin{cases}1-x & 0\leq x\leq 2\\ x-3 & 2\leq x\leq 4\end{cases}$$

$g(x)=fof(x)=f(f(x))$

$$g(x)=\begin{cases}x & 0\leq x<1\\ 4-x & 3\leq x<4\end{cases}$$

$(gogogog)(x)=g(g(g(g(x))))$

57. System of equation have atleast one solution

So $\Delta=\Delta_1=\Delta_2=\Delta_3=0$

$\Delta_2=a-b+c=0$

58. $a=4;b=\frac{1}{2}\Rightarrow x^3+x^2-ax-8b=0\Rightarrow x^3+x^2-4x-4=0$

Roots are $-1,-2,2$

$x^2+px+2p=0\Rightarrow\Delta>0\Rightarrow p<0(\text{or})\,p>8$

Put $x=-1\Rightarrow p=-1$, put $x=2\Rightarrow p=-1$, put $x=-2\Rightarrow 4$ to $=0(F)\Rightarrow p=-1$

59. $\sin x=X$

$$\lim_{X\to 0}\frac{\sin^{-1}X\sin X-X^2}{X^6\dfrac{\sin^{-6}X}{X^6}}$$

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$$=\frac{\left(X-\frac{X^3}{6!}+\frac{X^5}{5!}\right)\left(X+\frac{X^3}{6}+\frac{3X^5}{40}\right)}{X^6}-X^2$$

$$\frac{X^2+\frac{1}{6}X^4+\frac{3}{40}X^6-\frac{X^4}{6}-\frac{X^6}{36}+\frac{X^6}{120}-X^2}{X^6}=\frac{3}{40}-\frac{1}{36}+\frac{1}{120}=\frac{1}{18}$$

60. $\bar{r}\times\bar{b}=\bar{c}\times\bar{b}\Rightarrow\left(\bar{r}-\bar{c}\right)\times\bar{b}=0\Rightarrow\bar{r}-\bar{c},\bar{b}$ are collinear each

Let $\bar{r}-\bar{c}=\lambda\bar{b}\Rightarrow\frac{\bar{b}}{\bar{r}-\bar{c}}=\frac{1}{\lambda}\Rightarrow\frac{\left|\bar{b}\right|}{\left|\bar{r}-\bar{c}\right|}=2$

61. $A\left(3t+7,\,2t+5,\,t+3\right),B\left(2s+1,\,4s-1,\,3s-1\right)$

Distance $=\frac{2s-3t-6}{2}\pm\frac{4s-2t-6}{2}\pm\frac{3s-t-4}{1}$

$4s-2t-6=6s-2t-8\Rightarrow 2s=2\Rightarrow s=1$

$2s+t=0\,;\Rightarrow t=-2$

$\therefore A\left(1,1,1\right),B\left(3,3,2\right)$

$\therefore$ distance $=0$

62. $\frac{dy}{dx}\sin x-y\cos x+\frac{\sin^2 x}{x^2}=0\Rightarrow\frac{dy}{dx}-y\frac{\cos x}{\sin x}=\frac{-\sin x}{x^2}$

$\text{I.F}=e^{-\int\frac{\cos x}{\sin x}dx}=\frac{1}{\sin x}\Rightarrow\frac{y}{\sin x}=-\int\frac{1}{x^2}dx=\frac{1}{x}+c$

$x\to\alpha\Rightarrow y\to 1\Rightarrow c=0\Rightarrow y=\frac{\sin x}{x}\Rightarrow\underset{x\to 0}{Lt}\,f\left(x\right)=1$

63. Independent events

64. $g\left(x\right)$ is continuous and differentiable at $x=0$

So $\underset{x\to 0^-}{Lt}\,g\left(x\right)=\underset{x\to 0^+}{Lt}\,g\left(x\right)\Rightarrow\boxed{A=1}$

$$g'\left(x\right)=\left\{\begin{matrix}\frac{B}{\sqrt{1-x^2}} & x\geq 0\\ f'\left(x\right) & x<0\end{matrix}\right\}\Rightarrow\frac{B}{\sqrt{1-x^2}}=f'\left(x\right)\text{ at }x=0\Rightarrow\boxed{B=-1}$$

$A^3+B^3=0$

65. $\frac{x^2}{k^2a^2}+\frac{y^2}{a^2}=1\Rightarrow e=\sqrt{\frac{k^2a^2-a^2}{k^2a^2}}=\frac{\sqrt{k^2-1}}{k}$

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$\text{Foci} = a\sqrt{k^2-1} \rightarrow (1)$

$\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1 \Rightarrow e = \sqrt{2} \Rightarrow \text{Foci} = \sqrt{2}a \rightarrow (2)$

$\cancel{a}\sqrt{k^2-1} = \sqrt{2}\cancel{a} \Rightarrow k^2 - 1 = 2 \Rightarrow k^2 = 3 \Rightarrow k = \pm\sqrt{3}$

$(\alpha+\beta)^2 = 0$

66. $t_r = (r^2 - r + 1)r! = (r^2 + 2r + 1 - 3r)r!$

$= (r+1)^2 \cdot r! - 3r \cdot r! = (r+1)\cdot(r+1)! - 3r \cdot r!$

$= [(r+2)! - (r+1)!] - 3[(r+1)! - r!]$

$S = \sum_{r=1}^{50} [(r+2)! - (r+1)!] - 3[(r+1)! - r!]$

$= 52! - 2! - 3(51! - 1) = 1 + 52! - 3(51)! = 1 + 49 \cdot (51)!$

67. $Df \rightarrow -1 \le \frac{2-|x|}{4} \le 1 \Rightarrow x \in [-6, 6] \rightarrow (1)$

$3 - x \ne 1; 3 - x > 0 \Rightarrow x < 3 \Rightarrow x \in (-\infty, 3) - \{2\} \rightarrow (2)$

From (1) & (2) $\Rightarrow (-6, 2) \cup (2, 3)$

$(b, c) = (2, 3)$

$A = 2\sqrt{8}\int_2^3 \sqrt{x}\,dx = 4\sqrt{2}\frac{2}{3}\left[x\sqrt{x}\right]_2^3$

$A = \frac{8\sqrt{2}}{3}\left[3\sqrt{3} - 2\sqrt{2}\right] = 8\sqrt{6} - \frac{32}{3}$

68. $f(0) = 10$

$f'(0) = -2$

$f(0) + f'(0) = 8$

69. If satisfies $x = 1; y = -1, z = -1, w = 1 \Rightarrow x + y + z + w = 0$

70. $2\int_0^{\pi/4} \frac{24\sin x(1+\cos^2 x)}{(4 + 6\cos^2 x + \cos^6 x)\cos x}\,dx$

Let $\cos x = t$

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$$2\int_{1/\sqrt{2}}^{1}\frac{24\left(1+t^2\right)}{t\left(4+6t^2+t^6\right)}dt=-2\int_{1/\sqrt{2}}^{1}\frac{-24\left(t^{-7}+t^{-5}\right)}{1+6t^{-4}+4t^{-6}}$$

$$=-2\log\left(1+6t^{-4}+4t^{-6}\right)\Big|_{1/\sqrt{2}}^{1}=-2\left[\log 11-\log(57)\right]=2\log\left(\frac{57}{11}\right)$$

$\therefore a+b=68$

71. $\bar{a}\cdot\bar{c}=3|\bar{c}|\Rightarrow|\bar{a}||\bar{c}|\cos\left(\bar{a},\bar{c}\right)=3|\bar{c}|$

$\cos\left(\bar{a},\bar{c}\right)=1$

$\therefore\bar{c}=\lambda\bar{a}\Rightarrow\bar{a}\,\|\,\bar{c}$

$$\therefore\cos\theta=\left|\frac{2-4+1}{3\sqrt{6}}\right|=\frac{1}{3\sqrt{6}}$$

$$216\cos^2\theta=\frac{216}{9\times6}=4$$

72. $f\left(x+y\right)=2^x f\left(y\right)+4^y f\left(x\right)\rightarrow(1)$

$$\frac{f\left(x\right)}{4^x-2^x}=\frac{f\left(y\right)}{4^y-2^y}$$

Let $f\left(x\right)=k\left(4^x-2^x\right)\Rightarrow f'\left(x\right)=k\left(4^x-\log y-2^x\log x\right)\Rightarrow f'\left(0\right)=\log 2\Rightarrow k=1$

$\therefore f\left(x\right)=4^x-2^x\Rightarrow f\left(x\left(x-3\right)\right)=4^{x(x-3)}-2^{x(x-3)}$

$f\left(0\right)=0;f\left(1\right)=2$

$f\left(\left(3-x\right)x\right)=4^{(3-x)x}-2^{(3-x)x}$

$$I_1=\int_0^3 xf\left(x\left(x-3\right)\right)dx\Rightarrow I_1=\frac{3}{2}\int_0^3 f\left(x\left(x-3\right)\right)dx$$

$$I_2=\int_0^3 f\left(x\left(3-x\right)\right)dx$$

73. Let $z=x+iy$

$|z-2-2i|\le1$

$|x+iy-2-2i|\le1$

$|(x-2)+i(y-2)|\le1$

Now $|iz+2|_{\max}=|z-2i|$

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74. $L_1 : \frac{x-5}{0} = \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2 : \frac{x-\alpha}{0} = \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar

$$\Rightarrow \begin{vmatrix} 5-\alpha & 0 & 0 \\ 0 & 3-\alpha & -2 \\ 0 & -1 & 2-\alpha \end{vmatrix} = 0 \Rightarrow (5-\alpha)(\alpha^2 - 5\alpha + 4) = 0$$

$\Rightarrow \alpha = 5, 6 - 5\alpha + \alpha^2 - 2 = 0 \Rightarrow \alpha^2 - 5\alpha + 4 = 0 \Rightarrow \alpha = 1, 4$

$\therefore \alpha = 1, 4, 5 \Rightarrow \text{sum} = 10$

75. $f'(x) = e^{\frac{-x^2}{4}}(4 - x^2) \cdot 2x$

$f'(x) = 0 \Rightarrow x = 0, -2, 2$

$f''(x) = e^{x^2/4} \cdot \frac{2x}{4}(4 - x^2) \cdot 2x + e^{x^2/4}(-2x)(2x) + e^{x^2/4}(4 - x^2)2$

$f''(0) = \min \text{ at } x = 0$

$f''(2) = \max \text{ at } x = 2 \qquad p = 4 + 4 = 8$

$f''(-2) = \max \text{ at } x = 2 \qquad q = 0$

$p + 2q = 8 + 0 = 8$

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